

Jack Wesley Lipson

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Education

University of California, Berkeley Spring 2026

B.A. in Pure Math with Honors – Percy Lionel Davis Award (department)

Graduate coursework: **A+**: algebraic topology, Riemannian geometry. **A**: measure theory (2 semesters), differential topology, symplectic geometry, quantitative symplectic geometry, Lie groups/algebras, complex manifolds, number theory. **Pass (only option)**: non-commutative geometry, deep learning

Portola High School, Irvine, CA Spring 2022

Graduated summa cum laude 1 year early.

Work Experience / Research

University of California, Berkeley

Marco Troper REU (paid, 30 hrs/wk) Summer 2026

- *Ongoing: project on quasi-periodic research.*

Berkeley Math Circle Instructor (prior: TA) (paid, 6 hrs/wk) Fall 2025 - Present

- Hone problem-solving skills of advanced 5th and 6th graders.

Math Department Teaching Assistant (paid, 30 hrs/wk) Fall 2024 - Present

- Grade graduate algebraic topology + 2 x complex analysis + differential metric geometry (Fall '26)
- Grade number theory + complex analysis (Summer '26)
- Grade differential metric geometry + honors (and non-honors) complex analysis (Spring '26)
- Grade model theory + differential metric geometry + algebraic topology (Fall '25)
- Grade honors complex analysis + abstract algebra (Spring '25) + linear algebra (Summer '25)
- Grade graduate + undergrad differential geometry (Fall '24)

Upper Division Math Department Tutor (paid, 20 hrs/wk) Summer 2023 - Spring 2026

- Tutor drop-in for real + complex analysis, linear + abstract algebra.
- Create + facilitate 5 review sessions/semester in above classes, proof skills, and category theory.

Math/Physics Directed Reading Program (1-on-1 weekly with grad student) All 2024

- Presentation: Supersymmetric QM + Read: Atiyah-Macdonald, Hartshorne, some Atiyah papers.

Scale AI, San Francisco Summer 2025

Technical Advisor Intern - GenAI (paid, 20+ hrs/wk)

- Benchmarking for STEM research, biosecurity, and other projects.

Stanford University – Pre-Collegiate Studies

Stanford University Mathematics Camp Admissions Exam Reviewer (paid) Spring 2024 + 2025

- Assessed high school applicants' math maturity via personal essays + grades + proof-based exam.

AwesomeMath

Summer Program Instructor (prior: TA) (paid, 40 hrs/wk) Summer 2024 + 2025 + 2026

- Led discussion, held office hours, and graded HW in 17 classes for very advanced high schoolers.
- Courses taught competitive math skills in algebra, combinatorics, number theory, and geometry.

United States Air Force Fall 2022 - Spring 2024

Reserve Officers' Training Corps Cadet (paid)

- American Legion Cathay Post AFROTC Scholarship (only freshman recipient). Spring 2023

Skills / Interests

Technical: microfabrication, LaTeX, Python, C++, Docker, git, MATLAB, GIS, Fusion 360, Solidworks

Languages: English > Mandarin > French > Russian **Interests:** Cooking, bike maintenance, woodworking